

CARLA TY

COMPUTER ENGINEERING UNDERGRADUATE

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PROFESSIONAL SUMMARY

Innovative and detail-oriented Computer Engineering undergraduate with hands-on experience in image processing, embedded systems, IoT, and circuit design. Passionate about problem-solving and eager to apply technical skills in real-world applications. Seeking opportunities to grow in the tech industry while contributing to innovative solutions.

EXPERIENCE

CUSTOMER SERVICE REPRESENTATIVE EXPERIENCE (TELECOMMUNICATION ACCOUNT)

JUNE 2023 – SEPTEMBER 2023

Alorica | Cubao, Quezon City

Working in a team environment, I gained valuable experience in addressing customer inquiries, resolving concerns, and providing technical assistance to ensure customer satisfaction. I assisted with product activation, troubleshooting technical issues, and guiding customers in optimizing their devices and services. Additionally, I handled transactions with accuracy, including processing orders, payments, returns, and exchanges. By educating customers on product features and benefits, I helped them make informed decisions and maximize their experience with the products and services offered.

PROJECTS

IMAGE PROCESSING BAMBOO USING YOLO ALGORITHM (THESIS)

Image Processing: YOLOv5 Algorithm, Roboflow, Raspberry Pi 4B, Ubuntu OS

- Develop a device to distinguish between running and clumping bamboo growth habits using image data
- Utilize the YOLOv5 algorithm to automatically detect and classify bamboo growth habits patterns, allowing for efficient processing of image-based datasets
- Analyzed the performance of the created model using a confusion matrix
- Evaluate the practical applicability of the device and algorithm in real-world bamboo habitats

VENDING MACHINE USING ATMEGA328P (CAPSTONE)

ATMEGA328P: Raspberry pi 4B, PCB Design, Buzzer, Ultrasonic Sensor

- Developed a coin-operated printing vending machine using an ATmega328P microcontroller
- Implement a combinational circuit for dispensing change, returning 2 pesos when a 5-peso coin is inserted.
- Accepts 1-peso and 5-peso coins with a coin slot module, dispensing forms priced at 3 pesos each.
- Features a user interface with a keyboard for entering required information on the form before printing.
- Uses a 7-segment display for a synchronous countdown of pages printed, controlled via a JK flip-flop-based 4-bit down counter.
- Equipped with a sensor and buzzer to alert when paper supply is empty.

MINI SMART HOME SECURITY SYSTEM

EMBEDDED SYSTEM: ARDUINO UNO

- Buzzer: Alerts occupants to intrusions.
- Temperature sensor: Detects fire or extreme temperatures.
- LEDs: Light up when the main door opens.
- Stepper/servo motors: Automate curtains and door control.
- Keypad: Allows secure access with password entry.
- Ultrasonic sensor: Detects movement or obstacles, activating the buzzer.
- Integration: These components create a smart, secure, and automated home system

IOT WITH MINI SMART MEDICINE CABINET

EMBEDDED SYSTEM: ARDUINO UNO, ESP32, IoT ThinkSpeak (Cloud data collection)

- Integrates sensor technology for effective medication management and storage.
- Temperature sensor ensures optimal storage conditions, maintaining medication potency.
- Ultrasonic sensor tracks cabinet access by logging door openings and closings for usage monitoring and security.
- IR sensor monitors the quantity of medicines, helping users keep track of stock levels.
- Automated reminders through a buzzer and LED indicators alert users to low stock or if the cabinet is left open.
- Prioritizes medication adherence and safety, preventing errors and maintaining proper storage conditions.

MINI SMART BICYCLE FOR CYCLIST SAFETY

EMBEDDED SYSTEM: ARDUINO UNO

- Addresses the need for cyclist protection using ultrasonic sensors, LED indicators, a microcontroller Arduino Uno, LCD display, and a buzzer.
- Alerts cyclists about approaching vehicles from behind with a multi-faceted warning system.
- Incorporates a manual turning signal system for enhanced situational awareness.
- Features three buttons for signaling: left, right, and off.
 - Pressing left or right triggers corresponding blinking yellow LEDs to indicate direction.
 - The middle button turns off the signal.
- LCD display shows safety recommendations and messages to the cyclist.

WEATHER APPLICATION

SOFTWARE DESIGN: C#, VSC, MYSQL, DATABASE MANAGEMENT SYSTEM (DBMS), FIGMA

Designed and developed the user interface for the application, incorporating key features such as a Welcome Page, Sign Up/Log-In Page, Search Bar, Notification Tool, Location Pings, Menu Bar, Settings, and more to ensure a seamless user experience. Created a database diagram to outline the data structure and relationships. Leveraged a Database Management System (DBMS) to perform Create, Read, Update, and Delete (CRUD) operations, ensuring efficient data management and system functionality.

AWSCC-MAPUA OFFICAL WEBSITE (BACKEND DEVELOPER)

- AWS Service: API Gateway, Lambda, S3, DynamoDB, IAM
- A collaboration work with other members of AWSCC-MAPUA to make a website for AWS Club Mapua. Currently on-going.

CLOTHING APPLICATION'S USER-INTERFACE

SOFTWARE DESIGN: FIGMA

Designed and implemented a user-friendly interface for an application, integrating essential features such as a Welcome Page, Sign-Up/Log-In Page, Search Bar, Notification Tool, Payment Page, Delivery Page, Menu Bar, Settings, and more. The interface focuses on intuitive navigation and accessibility to enhance the overall user experience while maintaining functionality and aesthetic appeal.

SKILLS

Microcontrollers & Development Boards: Arduino UNO, ESP32, ESP8266, Raspberry Pi 4, ATMEGA328P

Embedded Programming: C/C++, Python, JavaScript

Hardware Design: PCB Design, Circuit Schematic Design, Sensor Integration

Communication Protocols: UART, I2C, SPI, MQTT

Firmware & Real-Time Systems: Embedded C, RTOS Basics,

Networking & Cloud Integration: AWS IoT, HTTP/HTTPS APIs, RESTful Services, DynamoDB, AWS Lambda, AWS IAM, AWS S3

Version Control & Development Tools: Git/GitHub, Linux CLI, VS Code, Anaconda, Jupyter Notebook, Docker, Arduino IDE

Simulation & Prototyping: Proteus, Tinkercad

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

AUGUST 2020 – DECEMBER 2025

SPECIALIZATION IN EMBEDDED SYSTEM TECHNOLOGY (ONGOING)

Mapua University, Intramuros Manila

- Non-Technical Team Member of the Mapua University's ICpEP (Institute of Computer Engineers of the Philippines)
 - Cloud Infrastructure team member of AWS Cloud Club (Backend Developer)
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CERTIFICATIONS

- **TECHNICAL SUPPORT FUNDAMENTALS**
GOOGLE
ISSUED MAR 2021
CREDENTIAL ID KUTMBU3FR863
 - **THE BITS AND BYTES OF COMPUTER NETWORKING**
COURSERA
ISSUED MAY 2021
CREDENTIAL ID
COURSERA.ORG/VERIFY/RMBC3MDRSZGL
 - **APPLIED PLOTTING, CHARTING & DATA REPRESENTATION IN PYTHON**
COURSERA
ISSUED OCTOBER 2022
CREDENTIAL ID
[HTTPS://COURSERA.ORG/VERIFY/B5LE56PFTAD2](https://COURSERA.ORG/VERIFY/B5LE56PFTAD2)
 - **INTRODUCTION TO GIT AND GITHUB**
COURSERA
ISSUED MAY 2021
CREDENTIAL ID
[HTTPS://COURSERA.ORG/VERIFY/HFP33FU4UHW4](https://COURSERA.ORG/VERIFY/HFP33FU4UHW4)
 - **SYSTEM ADMINISTRATION AND IT INFRASTRUCTURE SERVICES**
COURSERA
ISSUED OCTOBER 2022
CREDENTIAL ID
[HTTPS://COURSERA.ORG/VERIFY/TLW8W7B4WYK8](https://COURSERA.ORG/VERIFY/TLW8W7B4WYK8)
 - **OPERATING SYSTEMS AND YOU: BECOMING A POWER USER**
COURSERA
ISSUED APRIL 2022
CREDENTIAL ID
[HTTPS://COURSERA.ORG/VERIFY/9R2CTEQ7HKWT](https://COURSERA.ORG/VERIFY/9R2CTEQ7HKWT)
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